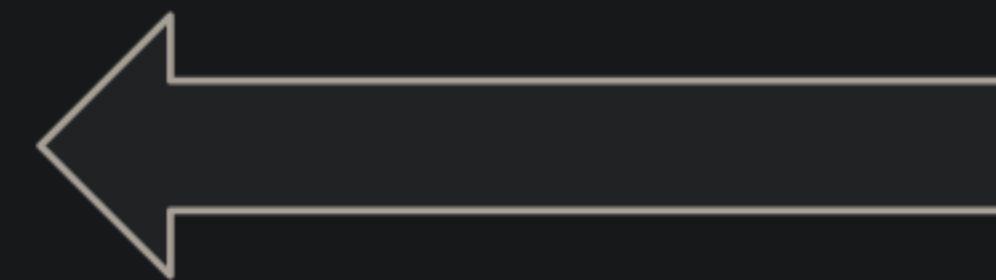


TASK CHECKLIST

- ☒ Use requirements elicitation techniques (interview, JAD session, questionnaire, document analysis, and observation).
- ☒ Apply requirements analysis strategies as needed to discover underlying requirements.
- ☒ Develop the requirements definition.
- ☐ Develop use cases.
- ☐ Develop data flow diagrams.
- ☐ Develop entity relationship model.
- ☐ Normalize entity relationship model.

Categories of Requirements

- Business Requirements
- **User Requirements**
- Functional requirements
- Non-functional requirements
- System requirements



Use cases are a way to express user requirements: **the things the user needs to accomplish with the new system**

Example

- As a student, what are some of **the things you can do in Learning Hub?**
 - Those *were* the User Requirements
- If I ask you to express those processes (User Requirements) in order to clearly communicate it to someone else, how would you do so?

A user-system dialog

- **Functional requirement:** "The kiosk system will provide the option to purchase postage stamps"
- Notice that a functional requirements has only **few details**
- **Developing the dialog** helps us create a *much more complete* description of functional requirements.

System:	Displays Welcome message and flashes "Start" button
User:	Presses "Start" button
System:	Displays list of options (buy stamps, mail package, etc.)
User:	Presses "Buy Stamps" button
System:	Displays list of stamp-buying options (type, denomination, quantity)
User:	Selects desired stamps by pressing button next to choice
System:	Displays dollar amount of purchase and asks for purchase confirmation
User:	Presses "Yes" button to confirm purchase
System:	Asks user to swipe card and flashes light next to card reader
User:	Swipes debit card in card reader
System:	Displays keypad for PIN entry and requests PIN
User:	Types in PIN on keypad
System:	Confirms payment transaction and asks if user wants a printed receipt
User:	Presses "Yes" button to request receipt
System:	Prints receipt and asks user to take receipt
User:	Takes receipt
System:	Displays message to wait while it prints the purchased stamps
User:	Takes stamps when printing is done
System:	Displays Thank You Message for 10 seconds, then displays Welcome message

“The kiosk system will provide the option to purchase postage stamps”

The kiosk system will:

- enable the user to begin a transaction
- display a list of purchase transaction options that are available on the kiosk
- enable the user to specify the type of purchase transaction desired
- display a list of options for the purchase type
- enable the user to specify the desired type of postage and quantity desired
- display the amount due for the purchase
- enable the user to confirm the purchase
- enable the user to pay for the purchase with credit/debit card
- complete the credit/debit card transaction securely
- print receipt if requested by user
- print stamps purchased

This is analysis!
Imagining the details of a functional requirement.

After you've developed a user-system dialogue, you can structure it as **a formal use case** which includes even **more details**.

The purpose of use cases

- Represents the **interaction** required between the user and the system (user-system dialog).
 - Users perform **activities**
 - System generates **responses**
- Provides an **external/functional view** of a process rather than the internal mechanism.
 - So they help us fully understand the **functional requirements**

Event-driven Modeling

Developing use cases are an example of **event-driven modeling**:

an event triggers a user-system dialog/interaction.

Creating Use Cases

1. Identifying the major use cases

- Look for things that happen (**events**) which cause activities to be performed.
- Try to discover the **major response(s)** to those events.
- develop the **event-response list**.

Each of these event-response items is a **major use-case** of the system.

Event	Response
1. New shop work order is received	• Shop work order arrival time is recorded
2. Shop work order is assigned to qualified technician	• Work order added to technician's work assignment
3. Parts Request for base model drone and additional components generated	• Parts request stored • Inventory department receives Parts Request
4. Requested parts arrive in Shop Parts room	• Parts arrival recorded on Parts Request
5. All parts on a Parts Request are available	• Technician notified that all parts are available
6. Technician records date/time work begins on work order	• Shop work order updated with start date/time
7. Technician records completion of work on work order	• Shop work order updated with completion date/time • Customer is notified of order completion

FIGURE 4-5 Sample event-response list.

2. Shop work order is assigned to qualified technician	• Work order added to technician's work assignment
3. Parts Request for base model drone and additional components generated	• Parts request stored • Inventory department receives Parts Request

After you identify the major use cases, **fill in the top parts of the use case form**: name, id, primary actor, description, and trigger.



Use Case Name: Create Parts Request	ID: UC-3	Priority: High
Actor: Shop Manager		
Description: This use case describes how the Shop Manager creates a Parts Request.		
Trigger: Shop manager receives notice of new shop work order arrival from Sales System.		
Type: ☒ External ☐ Temporal		
Preconditions: Shop manager is authenticated Parts Request application is available and online Inventory application is available and online		

Goal: just identify major use cases with **basic information** without jumping into one use case and describing it completely.



4. Requested parts arrive in Shop Parts room	• Parts arrival recorded on Parts Request
--	---

Use Case Name: Record Receipt of Component	ID: UC-4	Priority: High
Actor: Shop Parts Room Clerk		
Description: This use case describes how the Parts Room Clerk Records delivery of drone component from Inventory Department.		
Trigger: Drone component arrives in Parts Room from Inventory Department.		
Type: ☒ External ☐ Temporal		
Preconditions: Parts Room Clerk is authenticated Parts Request application is available and online		



5. All parts on a Parts Request are available	• Technician notified that all parts are available
---	--

Use Case Name: Finalize Parts Request	ID: UC-5	Priority: High
Actor: Parts Room Clerk		
Description: This use case describes how the Parts Room Clerk finalizes a Parts Request.		
Trigger: Notification received that all parts are available for a Parts Request.		
Type: ☒ External ☐ Temporal		
Preconditions: Parts Room Clerk is authenticated All components listed on Parts Request have arrived in Parts Room		

FIGURE 4-7 Major use cases with basic information.

Notice how steps 3, 4 and 5 are parts of an overall goal: **Obtaining all required components**



Event	Response
1. New shop work order is received	• Shop work order arrival time is recorded
2. Shop work order is assigned to qualified technician	• Work order added to technician's work assignment
3. Parts Request for base model drone and additional components generated	• Parts request stored • Inventory department receives Parts Request
4. Requested parts arrive in Shop Parts room	• Parts arrival recorded on Parts Request
5. All parts on a Parts Request are available	• Technician notified that all parts are available
6. Technician records date/time work begins on work order	• Shop work order updated with start date/time
7. Technician records completion of work on work order	• Shop work order updated with completion date/time • Customer is notified of order completion

FIGURE 4-5 Sample event-response list.

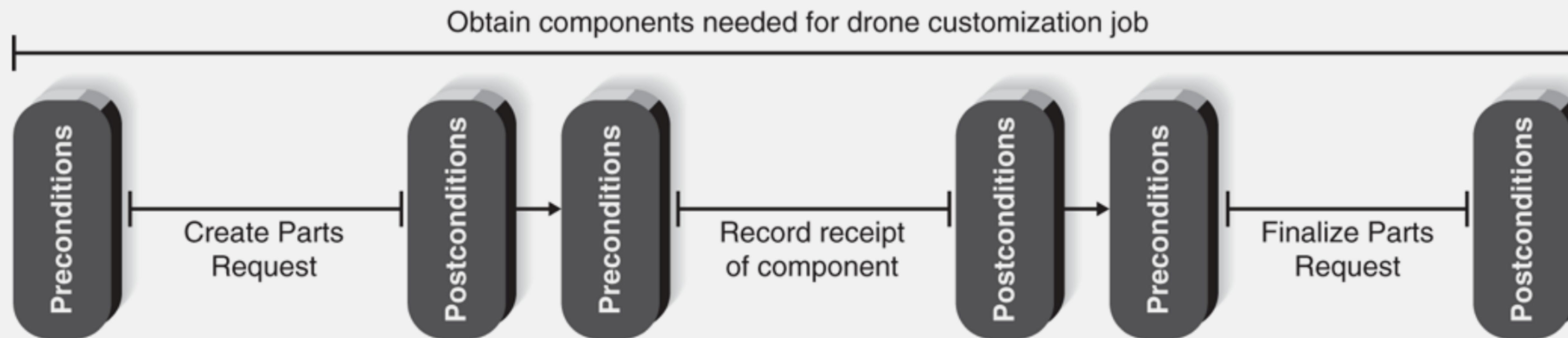


FIGURE 4-6 Chain of use cases for obtaining drone components.

2. Identify the major steps for each use case

- The users and analyst work together
- Ask the users: **what tasks need to be completed before the use case can begin?** → *Preconditions*
- The major steps are what an **independent observer** would see the user and system do in response to the event **when everything flows smoothly.**
- Each step should be the **same size** as others.
- Have the user visualize themselves performing the use case and write down the steps as if they were writing a **recipe for a cookbook.**
- Ask the users: **how do you know you are finished with a task?** → *Postconditions*

2. Shop work order is assigned to qualified technician	• Work order added to technician's work assignment
3. Parts Request for base model drone and additional components generated	• Parts request stored • Inventory department receives Parts Request



Use Case Name: Create Parts Request		ID: UC-3	Priority: High	...
Actor: Shop Manager				
Description: This use case describes how the Shop Manager creates a Parts Request				
Trigger: Shop Manager receives notice of new shop work order arrival from Sales System				
Type: ☒ External ☐ Temporal				
Preconditions: 1. Shop manager is authenticated 2. Parts Request application is available and online 3. Inventory application is available and online				
Normal Course: 1.0 Create Parts Request 1. Shop manager opens the shop work order 2. Shop manager opens a blank Parts Request 3. For each required component part a. Shop manager enters component part and quantity needed b. System queries the inventory datastore and records part status: (In stock/ Out of stock with expected date available) 4. Shop manager verifies Parts Request is complete 5. System stores new Parts Request 6. System transmits notice of Parts Request to be filled to Inventory Department			Information for Steps	
Postconditions: 1. New Parts Request record created and stored 2. Inventory Department notified of Parts Request to be filled				

3. Identify elements within steps

Goal: to identify major inputs and outputs for each step.

- The users and analyst begin **tracing the flow of the steps**: what **inputs** are used by each step? What **outputs** does it produce?
- You might discover **new steps** you missed before.
- Once the team is satisfied with steps, inputs and outputs, then they complete the **summary area**.

Use Case Name: Create Parts Request		ID: UC-3	Priority: High	...
Actor: Shop Manager				
Description: This use case describes how the Shop Manager creates a Parts Request				
Trigger: Shop Manager receives notice of new shop work order arrival from Sales System				
Type: ☒ External ☐ Temporal				
Preconditions: 1. Shop manager is authenticated 2. Parts Request application is available and online 3. Inventory application is available and online				
Normal Course: 1.0 Create Parts Request 1. Shop manager opens the shop work order 2. Shop manager opens a blank Parts Request 3. For each required component part a. Shop manager enters component part and quantity needed b. System queries the inventory datastore and records part status: (In stock/ Out of stock with expected date available) 4. Shop manager verifies Parts Request is complete 5. System stores new Parts Request 6. System transmits notice of Parts Request to be filled to Inventory Department			Information for Steps input: Shop work order ID output: Shop work order	
			input: Part ID and quantity needed output: Part inventory status	
			input: Part request status output: New part request record output: Parts Request to fill	
Postconditions: 1. New Parts Request record created and stored 2. Inventory Department notified of Parts Request to be filled				

Use Case Name: Create Parts Request		ID: UC-3	Priority: High
Actor: Shop Manager			
Description: This use case describes how the Shop Manager creates a Parts Request			
Trigger: Shop Manager receives notice of new shop work order arrival from Sales System			
Type: ☒ External ☐ Temporal			
Preconditions: 1. Shop manager is authenticated 2. Parts Request application is available and on-line 3. Inventory application is available and on-line			
Normal Course: 1.0 Create Parts Request 1. Shop manager retrieves the shop work order 2. Shop manager opens a blank Parts Request 3. For each required component part a. Shop manager enters component part and quantity needed b. System queries the inventory datastore and records part status: (In stock/ Out of stock with expected date available) 4. Shop manager verifies Parts Request is complete 5. System stores new Parts Request 6. System transmits notice of Parts Request to be filled to Inventory Department		Information for Steps input: Shop work order ID output: Shop work order input: Part ID and quantity needed output: Part inventory status input: Part request status output: New part request record output: Parts Request to fill	
Postconditions: 1. New Parts Request record created and stored 2. Inventory Department notified of Parts Request to be filled			
Summary Inputs	Source	Summary Outputs	Destination
Shop work order ID Part ID and quantity Part request status	Shop manager Shop manager Shop manager	Shop work order Part inventory status New part request record Parts Request to fill	Shop manager Inventory system Parts Request datastore Inventory department

4. Confirm the use case

- It is **users** who should confirm the use case is correct as written
- Ask the user to **role-play**.

5. Revise functional requirements

- Use cases help us understand the system to be developed.
- **New understanding** → modify functional requirements to provide insight to the development team on some **back-end processing** that will be needed that may not be obvious from the use cases alone.

Elements of a use case

Basic Information:

- Name & number
- Description
- Priority
- Actor
- Trigger

Use Case Name: Create preliminary custom drone order	ID: UC-6	Priority: High
Actor: Customer		
Description: The customer selects and customizes a commercial drone to purchase		
Trigger: Customer wants to purchase a commercial drone		
Type: ☒ External ☐ Temporal		
Preconditions: 1. The customer is authenticated by logging in to his account 2. The Sales System Order Processing application is online		
Normal Course: 1.0 Order a customized drone 1. The customer selects a base model drone from a list of models 2. The system provides availability status for that model (in stock, out of stock) 3. For out of stock status, system displays expected date available a. Customer accepts future availability date; proceed to step 4 b. Customer rejects future availability date; return to step 1 4. The system displays a list of options and upgrades for the selected model 5. The customer selects desired model options and upgrades 6. Preliminary order with cost estimate is created and displayed 7. Customer may return to step 4, confirm order, save for future consideration, or exit without saving 8. Unconfirmed orders are stored in Unconfirmed Custom Order datastore 9. Confirmed orders are saved in Confirmed Custom Order datastore 10. Shop manager is notified of Confirmed Order requiring approval		
Postconditions: 1. Unconfirmed order is stored in Unconfirmed Custom Order datastore 2. Confirmed order is stored in Confirmed Custom Order datastore 3. Shop manager sent notice of Confirmed Order requiring approval		

FIGURE 4-1 Create preliminary custom drone order use case—casual format.

Basic Information

- **Actor** can be a **person**, **another software system**, or a **hardware device** that interacts with the system.
 - Some organizations use the term **user role** because there may be several different user groups who **interact with the system in the same way**.
- **Trigger** can be an **external trigger** or a **temporal trigger**
- Priority is specially useful with a methodology that implements the system in a series of versions so the most essential features are targeted first.

Preconditions

Define the state the system must be in before the use case commences.

Use Case Name: Create preliminary custom drone order	ID: UC-6	Priority: High
Actor: Customer		
Description: The customer selects and customizes a commercial drone to purchase		
Trigger: Customer wants to purchase a commercial drone		
Type: ☒ External ☐ Temporal		
Preconditions: 1. The customer is authenticated by logging in to his account 2. The Sales System Order Processing application is online		
Normal Course: 1.0 Order a customized drone 1. The customer selects a base model drone from a list of models 2. The system provides availability status for that model (in stock, out of stock) 3. For out of stock status, system displays expected date available a. Customer accepts future availability date; proceed to step 4 b. Customer rejects future availability date; return to step 1 4. The system displays a list of options and upgrades for the selected model 5. The customer selects desired model options and upgrades 6. Preliminary order with cost estimate is created and displayed 7. Customer may return to step 4, confirm order, save for future consideration, or exit without saving 8. Unconfirmed orders are stored in Unconfirmed Custom Order datastore 9. Confirmed orders are saved in Confirmed Custom Order datastore 10. Shop manager is notified of Confirmed Order requiring approval		
Postconditions: 1. Unconfirmed order is stored in Unconfirmed Custom Order datastore 2. Confirmed order is stored in Confirmed Custom Order datastore 3. Shop manager sent notice of Confirmed Order requiring approval		

FIGURE 4-1 Create preliminary custom drone order use case—casual format.

Normal Course

- The **major steps** that are performed
- Sometimes called the “**happy path**”
- An external “**bird’s-eye**” perspective
- Might include **branching logical conditions**.

Use Case Name: Create preliminary custom drone order	ID: UC-6	Priority: High
Actor: Customer		
Description: The customer selects and customizes a commercial drone to purchase		
Trigger: Customer wants to purchase a commercial drone		
Type: ☒ External ☐ Temporal		
Preconditions: 1. The customer is authenticated by logging in to his account 2. The Sales System Order Processing application is online		
Normal Course: 1.0 Order a customized drone 1. The customer selects a base model drone from a list of models 2. The system provides availability status for that model (in stock, out of stock) 3. For out of stock status, system displays expected date available a. Customer accepts future availability date; proceed to step 4 b. Customer rejects future availability date; return to step 1 4. The system displays a list of options and upgrades for the selected model 5. The customer selects desired model options and upgrades 6. Preliminary order with cost estimate is created and displayed 7. Customer may return to step 4, confirm order, save for future consideration, or exit without saving 8. Unconfirmed orders are stored in Unconfirmed Custom Order datastore 9. Confirmed orders are saved in Confirmed Custom Order datastore 10. Shop manager is notified of Confirmed Order requiring approval		
Postconditions: 1. Unconfirmed order is stored in Unconfirmed Custom Order datastore 2. Confirmed order is stored in Confirmed Custom Order datastore 3. Shop manager sent notice of Confirmed Order requiring approval		

FIGURE 4-1 Create preliminary custom drone order use case—casual format.

Postconditions

- The final products of this use case
- Also serves to define the preconditions for the next use case in the series.

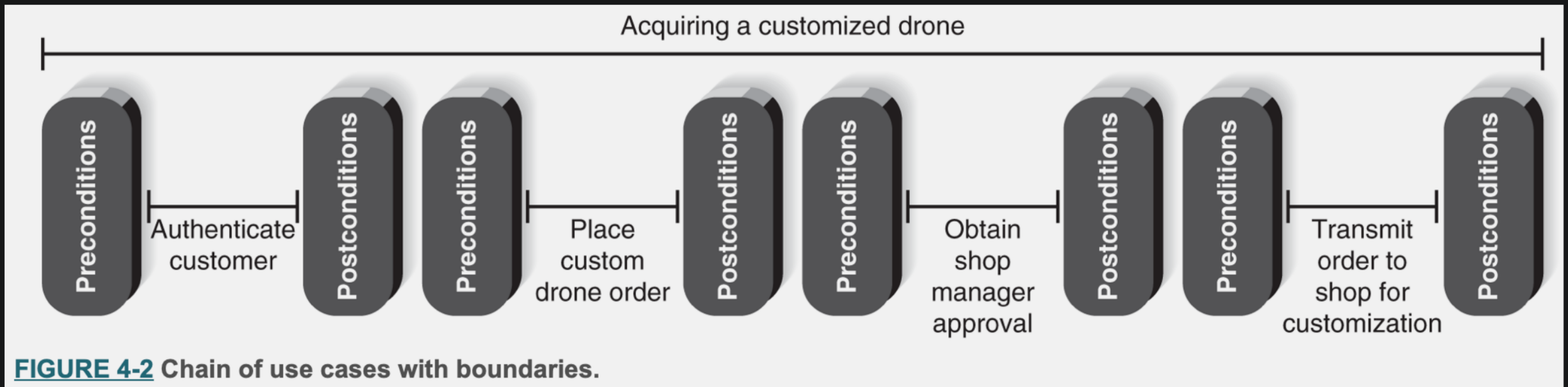
Use Case Name: Create preliminary custom drone order	ID: UC-6	Priority: High
Actor: Customer		
Description: The customer selects and customizes a commercial drone to purchase		
Trigger: Customer wants to purchase a commercial drone		
Type: ☒ External ☐ Temporal		
Preconditions: 1. The customer is authenticated by logging in to his account 2. The Sales System Order Processing application is online		
Normal Course: 1.0 Order a customized drone 1. The customer selects a base model drone from a list of models 2. The system provides availability status for that model (in stock, out of stock) 3. For out of stock status, system displays expected date available a. Customer accepts future availability date; proceed to step 4 b. Customer rejects future availability date; return to step 1 4. The system displays a list of options and upgrades for the selected model 5. The customer selects desired model options and upgrades 6. Preliminary order with cost estimate is created and displayed 7. Customer may return to step 4, confirm order, save for future consideration, or exit without saving 8. Unconfirmed orders are stored in Unconfirmed Custom Order datastore 9. Confirmed orders are saved in Confirmed Custom Order datastore 10. Shop manager is notified of Confirmed Order requiring approval		
Postconditions: 1. Unconfirmed order is stored in Unconfirmed Custom Order datastore 2. Confirmed order is stored in Confirmed Custom Order datastore 3. Shop manager sent notice of Confirmed Order requiring approval		

FIGURE 4-1 Create preliminary custom drone order use case—casual format.

Use cases are often performed in a sequence to accomplish an overall task.

Use cases in sequence

- Create smaller, more focused use cases **breaking down the whole process** down into parts.
 - Each use case is **less complex** and does not become confusingly large
 - We can potentially **reuse** a use case in other processes. (E.g. authenticate the customer)



Other sections

It may be helpful to include the following sections although not all use cases include them:

- Exceptions: Any error conditions or exceptions that may occur
- Alternative paths
- Summary of inputs and outputs
- Frequency of use
- Business rules
- Special requirements
- Assumptions
- Notes and issues

A few notes

- Use cases depict the user-system interactions as **abstract, technology-independent** steps.
- The process of building use cases is one of **gradual refinement**.
- Use cases are read and **used by two quite different groups of people**:
 - user/business expert
 - development expert
- Use cases only convey the **user's point of view**. The analyst transforms the user's point of view into the **developer's view** by creating functional requirements.
- Use cases and the functional requirements derived from them are used to create **test plans**.

To gather information for the use cases, use the same requirements determination techniques discussed in module 3, specially interviews and JAD sessions.

Step	Activities	Typical Questions Asked ^a
1. Identify the use cases. 	Start a use case report form for each use case by filling in the name, description, and trigger. If there are more than nine use cases, group them into packages.	Ask who, what, when, and where about the use cases (or tasks). What are the major tasks that are performed? What triggers this task? What tells you to perform this task?
2. Identify the major steps within each use case. 	For each use case, fill in the major steps needed to complete the task.	Ask how about each use case. What information/forms/reports do you need to perform this task? Who gives you these information/forms/reports? What information/forms/report does this produce and where do they go? How do you produce this report? How do you change the information on the report? How do you process forms? What tools do you use to do this step (e.g., paper, e-mail, phone)?
3. Identify elements within steps. 	For each step, identify its triggers and its inputs and outputs.	Ask how about each step. How does the person know when to perform this step? What forms/reports/data does this step produce? What forms/reports/data does this step need? What happens when this form/report/data is not available?
4. Confirm the use case. 	For each use case, validate that it is correct and complete.	Ask the user to execute the process, using the written steps in the use case—that is, have the user role-play the use case.

^a We have used the typical questions for the as-is model (e.g., “What are the ...”). These same questions can be used for the to-be model, but they would be phrased in the future tense (e.g., “What should be the ...”).

FIGURE 4-4 Steps for writing for use cases.